

AIMO Proof Pilot — Final Grade

Problem (#_ID): 1_DIVALL

Submission: team_3

Scoring scale (0/1/6/7) — two binary decisions:

Step 1. Is the solution essentially complete?

- If **no**: award **1** for significant partial progress, otherwise **0**.
- If **yes**: award **7**, or **6** if there is a minor error or omission.

Grade against the **markscheme** (docs/markschemes.pdf).

1. Final mark

Final mark (choose one):

2. Reasoning

Justify the mark with explicit reference to the markscheme points (e.g. ME1) where possible.

A complete solution along the standard milestones: it proves the a_i share a common residue $r \pmod{g}$ with $0 < r < g$ (establishing $\gcd(g, a_t) = 1$, hence $g \mid a_j - a_i$, so NEC1/NPP4 are met), reduces the $k = 3$ case to $r^2 + 1 = hg$, rules out $k \geq 4$ via $g < 500000$, and gives the construction $r = 1413$, $g = 998285$.

The single deduction is in the $k = 3$ tail analysis (around line 89): the script asserts that for $h \geq 5$ “the bound on r becomes even tighter and the maximum possible g is even smaller”. The *conclusion* that $h = 2$ is optimal is correct, but the stated reason is false — the bound on r in fact *loosens* as h grows (a larger h shrinks the $(r^2 + 1)/h$ term, admitting a larger r). It is the division by h , not a tighter bound on r , that drives g down. The two plots below illustrate this: the maximum admissible r increases with h , while the maximum $g = 10^6 - r$ correspondingly decreases. Since the final conclusion is nonetheless correct and the gap is easily repaired, this is a minor error (ME-level) rather than a major omission, so the solution scores **6** rather than **7**.

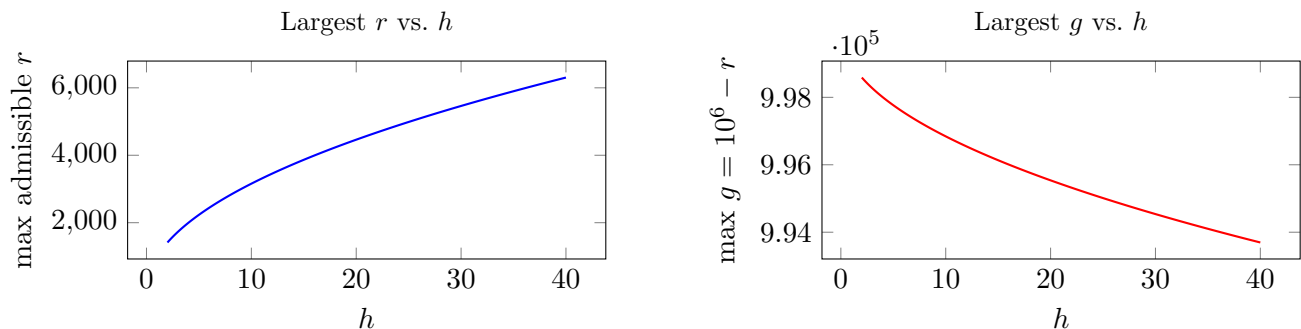


Figure 1: Maximising subject to $r + \frac{r^2+1}{h} < 10^6$. The largest admissible r (left) *increases* with h , while the corresponding $g = \frac{r^2+1}{h} = 10^6 - r$ (right) *decreases*. This contradicts the script’s claim that for $h \geq 5$ the bound on r becomes tighter.